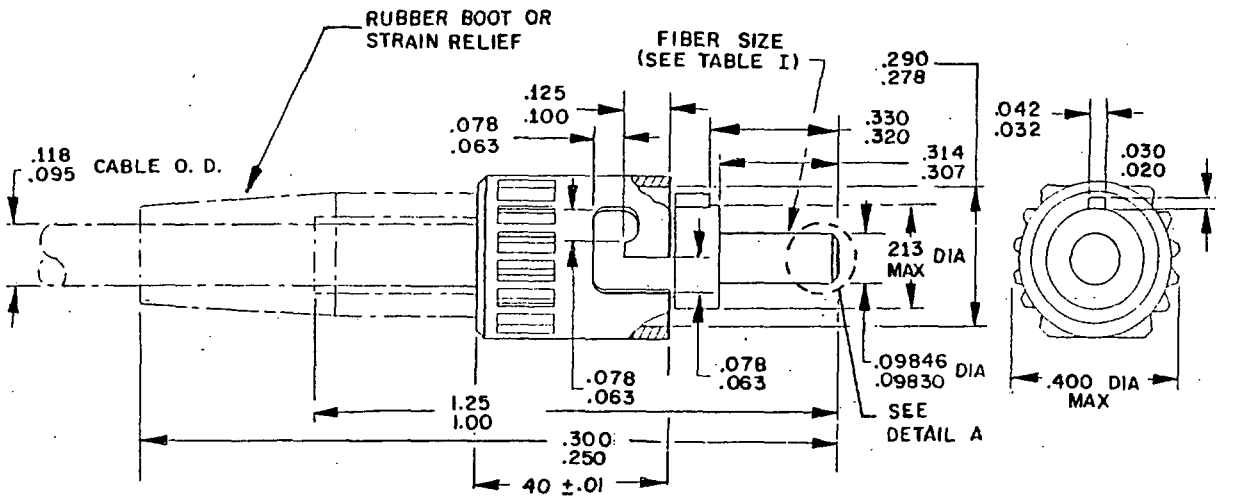
MILITARY SPECIFICATION SHEET

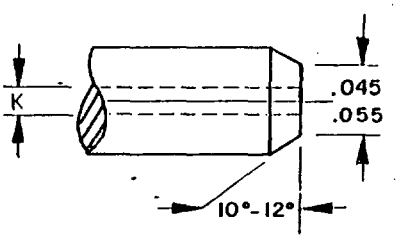
CONNECTOR, FIBER OPTIC, PLUG-RECEPTACLE-ADAPTER STYLE, 2.5 MM BAYONET CONNECTOR, PLUG, EPOXY, FOR 50/125 µm, 62.5/125 µm and 100/140 µm

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-83522.



Inches	mm	Inches	mm	Inches	mm
.01	0.3	.095	2.41	.290	7.36
.020	0.51	.09830	2.4968	.307	7.80
.030	0.76	.09846	3.5009	.314	7.98
.032	0.81	.100	2.54	.320	8.13
.042	1.07	.118	3.00	.330	8.38
.045	1.14	.125	3.18	.400	10.16
.055	1.39	.213	5.41	.40	10.2
.063	1.60	.250	6.35	1.00	25.40
.078	1.98	.278	7.06	1.25	31.8
				3.00	76.2



DETAIL A

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. K dimension is specified in µm. This dimension is critical to fiber alignment. Dimensions for fiber optic fiber shall be in accordance with DOD-F-49291.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Metals: The plug housing shall be of corrosion-resistant steel in accordance with QQ-S-763, class 303, /316; nickel-plated brass in accordance with QQ-B-626; nickel-plated zinc die cast alloy; or 6061 nickel-plated aluminum. Other metal components shall be of corrosion-resistant steel in accordance with QQ-S-763, class 303, /316; nickel-plated beryllium copper in accordance with QQ-B-530; nickel-plated brass in accordance with QQ-B-626, or 6061 nickel-plated aluminum.

Dust covers: Dust covers shall be thermoplastic.

Epoxies: Use Trabond 230, Eccobond 144B, Hysol 0141, or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements:

Cable diameter: In accordance with DOD-C-85045/2.

Cable configuration: In accordance with DOD-C-85045/2.

Fiber diameter: In accordance with DOD-F-49291/1 through DOD-F-49291/4.

Fiber optic contact:

Method of optical alignment: Sleeve/ferrule (bayonet).

Lens configuration: Not applicable.

Coating requirements: Not applicable.

Optical requirements:

Number of optical terminus: One.

Coupling loss (attenuation): 1 dB maximum per mated pair.

Weight: 4 grams maximum.

Polarization: Key/slot (bayonet).

Safety wire holes: Not required.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: Not applicable.

Coupling proof torque: Not applicable.

Coupling mechanism retention force: The force shall be 20 pounds minimum and shall not result in any physical damage.

Mating durability: In accordance with MIL-C-83522, mating durability paragraphs, except 500 cycles instead of 200 cycles.

Tensile load: In accordance with MIL-C-83522, tensile load paragraphs, except 9 kilograms instead of 18 kilograms.

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Cable retention: The load shall be 50 pounds minimum and shall not result in any physical damage. Optical properties shall be monitored before and after load. The load shall not be applied to the coupling mechanism.

Optical decoupling: The force required to optically decouple the connector shall be one pound minimum.

Submersion: Not applicable.

Ozone exposure: Not applicable.

Military part numbers: See table I.

TABLE I. Part numbers.

Part numbers M83522/12-	Fiber size	K dimension
01	50/125, 62.5/125	132.0 max. 127.5 min.
02	100/140	147.5 max. 142.5 min.
03	50/125, 62.5/125 1/	132.0 max. 127.5 min.
04	100/140 1/	147.5 max. 142.5 min.

1/ Salt spray, fluid immersion not applicable.

Plug accessories: Each plug shall be supplied with a minimum of one crimp ferrule; one protective cable-end cap; one 1.5 inch length of heat-shrink tubing or one rubber boot; one set of instructions; and listing of the epoxies specified in this specification sheet.

Note: The bayonet connector mates with MIL-C-83522/13 adapter bayonet and MIL-C-83522/14 PC mount.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0072-1)

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